D106-D107 | A price change is not causal proof

PRIVATE SYNTHETIC one-day onion market and separate one-hour ride-service case. Not actual Indian prices, transport contracts, platform accounting or policy authority.

Attempt before reference	Rebuild the inverse curves, resource-cost and ceiling benchmarks, then independently compute segmented midpoint arcs and contribution. Keep thousand rupees/day and rupees/hour separate. Actual errors, confidence and delayed transfer remain blank.
D106 | Market and cost	Demand P=80-Q; old supply P=20+Q. Q thousand kg/day; P rupees/kg, areas thousand rupees/day. Old Q 30/P 50/total 900; resource transport cost 20 shifts supply to 40+Q, Q 20/P 60/total 400.
D106 | Ceiling and recovery	Ceiling 50 under shock gives Qd 30/Qs 10/shortage 20. Highest-WTP allocation of 10 gives CS 250/PS 50/total 300, loss 100 versus feasible shocked 400. Allocation and queue costs are assumptions, not observations. Independent partial recovery cost 10 gives Q 25/P 55/total 625.
D107 | Segments and arcs	Fare 200->220; A 1000->900, B 1000->600. Signed midpoint arcs A -21/19, B -21/4, total -3. Total is midpoint-quantity weighted, not simple mean. An arc is descriptive of supplied points, not a constant or causal estimate.
D107 | Cost and decision	Toy operator keeps full fare, cost 150/ride. Old contribution 100000; new before policy 105000, after policy 8000 ->97000. Break-even 1542.857143 expected trips/1543 whole. New-policy-only cost 160 ->82000, minus 18000 versus unchanged old baseline.
Limits and change	No real shipment, fare, dispatch, participant, data disclosure, test, spending or public advice authorized. Static reference prose is Case-v1; revise it after input changes. Native arithmetic/curve drawing cannot establish actual demand, market rules, participant rights or mastery.

D106 inputs | Real resource cost, not a transfer

P=a-bQ and supply P=c+dQ. Q thousand kg/day; P rupees/kg; totals thousand rupees/day. Cost increments are real resource use, not taxes or carrier profit.

Demand intercept a, rupees/kg	80.00
Demand slope b, rupees/kg per thousand kg/day	1.00
Original supply intercept c, rupees/kg	20.00
Supply slope d	1.00
Shock extra resource cost per kg	20.00
Independent recovery extra resource cost per kg	10.00
Hypothetical ceiling under shock, rupees/kg	50.00

Core market model valid? (1=yes)	1.000000
Binding ceiling within model? (1=yes)	1.000000

Structural validity only. Unknown real causality, welfare distribution, quantity feasibility and action authority remain unknown. Negative computed contribution and decision changes stay visible. Inputs are fictional, not current market evidence.

D106 welfare | Compare feasible cost benchmarks

Competitive private-surplus model, no externality/market power/inventory/tax/demand shift. A cost shock changes the feasible benchmark. CS/PS are not accounting cash/profit.

Measure	Original	Shock	Recovery
Delivered supply intercept	20.000000	40.000000	30.000000
Equilibrium Q, thousand kg/day	30.000000	20.000000	25.000000
Equilibrium P, rupees/kg	50.000000	60.000000	55.000000
Consumer surplus, thousand rupees/day	450.000000	200.000000	312.500000
Producer surplus, thousand rupees/day	450.000000	200.000000	312.500000
Total private surplus	900.000000	400.000000	625.000000
Change versus old total (signed)	0.000000	-500.000000	-275.000000
Added resource cost on shock traded units	400.000000		
Lost old-cost gains on forgone units	100.000000		
Old-to-shock decomposition residual, 0	0.000000		
Resource saving on former shocked units	200.000000		
Restored gains on extra recovery units	25.000000		
Recovery decomposition residual, 0	0.000000		

Baseline loss 500 = 400 extra resource cost + 100 lost old-cost gains, not all policy DWL. Independent recovery gain 225 = 200 resource saving + 25 restored gains. Decomposition rows require recovery cost<=shock cost; otherwise rebuild the direction/basis explicitly, not a silent relabel.

D106 ceiling | Quantity is not an allocation rule

Ceiling price applies only under shocked supply. Highest willingness-to-pay buyers receive all supplied units; no queues/evasion/search costs. Random rationing may change CS.

Quantity demanded at ceiling	30.000000
Quantity supplied and assumed traded	10.000000
Shortage Qd-Qs	20.000000

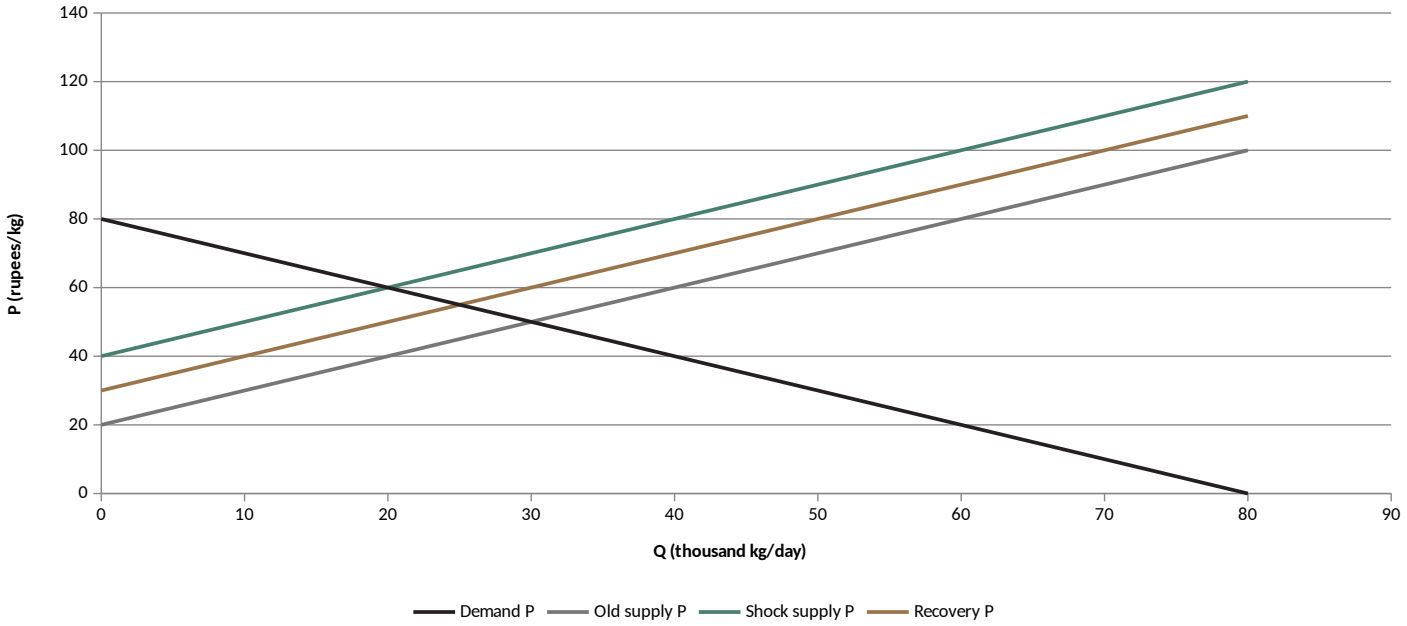
CS with highest-WTP allocation	250.000000
PS under shocked supply	50.000000
Ceiling total surplus	300.000000

Loss vs shocked efficient total	100.000000
Loss triangle, same new-cost benchmark	100.000000
Ceiling loss identity residual, 0	0.000000

Qd 30/Qs 10/shortage 20 does not mean 30 traded. CS 250/PS 50/total 300/loss 100 is conditional on efficient allocation and no extra rationing costs. Neither current Indian price-control law nor an actual allocation rule is established. Outside strict binding interior range this sheet deliberately blocks outputs.

D106 curves | Synthetic inverse curves

Quantity horizontal (thousand kg/day); price vertical (rupees/kg). Generated from fictional inputs, not a source figure or market observation. Ceiling allocation caveats on Ceiling sheet.



The curve intersection changes because cost shifts supply, not because higher price shifts demand. Curve chart is a model drawing; it does not identify transport-only causality. Demand/harvest/income/expectations evidence could contradict it. Ceiling output/areas are shown separately to avoid claiming a rationing allocation from the lines.

D107 inputs | Segment responses and avoidable costs

Independent one-hour toy operator keeps FULL fare and pays stated variable costs, not a real platform accounting model. Fixed costs unchanged and omitted from both; no taxes, capex, financing or working capital.

Old fare, rupees/trip	200.00
New fare	220.00
Old A completed trips	1000.00
New A completed trips	900.00
Old B completed trips	1000.00
New B completed trips	600.00
Old and base-new variable cost / trip	150.00
Avoidable new-policy hourly cost	8000.00
Separate downside NEW-policy variable cost	160.00

Ride contribution inputs valid? (1=yes)	1.000000
Midpoint arc defined? (1=yes)	1.000000

Structural validity only. Unknown real causality, welfare distribution, quantity feasibility and action authority remain unknown. Negative computed contribution and decision changes stay visible. Inputs are fictional, not current market evidence.

D107 elasticity | Match midpoint denominators

Signed arcs from two stipulated fare/quantity points. Not causal estimates or constants for extrapolation. Quantity-weight reconciliation differs from simple average.

Measure	A	B	Total
Old quantity	1000.000000	1000.000000	2000.000000
New quantity	900.000000	600.000000	1500.000000
Midpoint quantity	950.000000	800.000000	1750.000000
Quantity change / midpoint	-10.5263%	-50.0000%	-28.5714%
Fare change / midpoint	9.5238%	9.5238%	9.5238%
Signed midpoint elasticity	-1.105263	-5.250000	-3.000000
Reported magnitude	1.105263	5.250000	3.000000
Midpoint quantity weight	54.2857%	45.7143%	100.0000%
Weighted signed segment arcs	-3.000000		
Weighted-total residual, 0	0.000000		
Simple mean signed arcs, NOT total	-3.177632		

A -1.105263/B -5.25/total -3; midpoint weights 950/1750 and 800/1750. Completed trips may be supply constrained; wait times, weather, rider mix or service changes can confound observed points. No actual randomized or other identification design was performed or authorized.

D107 contribution | Revenue and margin can disagree

Rupees for the same one-hour comparison. New policy cost applies once to NEW policies only; downside 160 does not change old cost 150. Expected-trip thresholds and whole-trip ceilings differ.

Measure	Old baseline	New policy	New-only cost downside
Total completed trips	2000.000000	1500.000000	1500.000000
Fare / trip	200.000000	220.000000	220.000000
Variable cost / trip	150.000000	150.000000	160.000000
Revenue	400000.000000	330000.000000	330000.000000
Variable expense	300000.000000	225000.000000	240000.000000
Contribution before policy	100000.000000	105000.000000	90000.000000
Avoidable policy cost	0.000000	8000.000000	8000.000000
Contribution after policy	100000.000000	97000.000000	82000.000000
Change vs old contribution	0.000000	-3000.000000	-18000.000000

Expected trips to match old contribution	1542.857143
Minimum WHOLE trips to match old	1543.000000
Expected B trips if A stays new A	642.857143
Minimum WHOLE B trips with A fixed	643.000000

Old 100000/new 105000 before policy/new 97000 after 8000: minus 3000. Threshold 1542.857143, whole 1543; with A 900, B 642.857143/whole 643. New-only cost 160 -> 82000/minus 18000. Threshold is conditional arithmetic, not evidence demand is available at no extra cost or approval to change fare.

Audit | Identities are not market evidence

Synthetic arithmetic checks, not independent human review, real rationing policy, demand evidence or permission. A broken total-surplus reference blocks affected summary outputs.

Core welfare identities valid? (1=yes)	1.000000
Shocked total, guarded by welfare audit	400.000000
Ride bridge and weighted arcs valid? (1=yes)	1.000000

An undefined midpoint arc at unchanged fare is not zero elasticity; contribution remains independently computable. No positive margin or valid arithmetic gate authorizes actual price controls, fares, shipping, dispatch, experiments, data disclosure, contact or spending.

Evidence | Keep assumptions separate from observations

Original synthetic D106/D107 exercises at https://top1os.dev/. No new empirical source reading/full audiovisual review claimed. Existing selected teaching links remain in original course.

D106 | Evidence index	Market inputs -> Welfare/Ceiling/Curves -> Audit. One day, Q thousand kg/day, price rupees/kg, money thousand rupees/day. Transport increment is real resource use with no tax/carrier recipient. Ceiling CS depends on highest-WTP allocation and no rationing costs.
D107 | Evidence index	Ride inputs -> Elasticity/Contribution -> Audit. One hour, completed trips, rupees/trip and rupees totals. Full-fare toy operator accounting is stipulated; no actual platform take rate. Separate new-only cost downside; old baseline stays unchanged.
Strongest authored objection	Transport may not be the only shock and completed rides may be supply limited. Response: preserve source/input/identification uncertainty. Test coherent alternatives without pretending this synthetic model has estimated demand or validated any policy.
Day 0 | Scope and evidence roles	Fictional market/transport and demand/cost/safety roles verify units, period, matched services, alternative causes and allowed data use before any affected real work. These names are not accepted owners or actual participant permission.
Day 1/7 | Review and stop	Check original claim, contrary evidence, affected cells, conditional decision and strongest actual objection only if obtained. Stop affected real work without actual authority and valid evidence. No actual price control, fare experiment, participant, shipment or public claim authorized.
Delayed transfer	Without notes reduce resource shock 20 to 10 while choose a binding interior ceiling explicitly; separately change B response and new-policy cost. Recompute welfare benchmark, allocation assumptions, midpoint weights and contribution. Record actual reasoning/errors and delayed novel result only if performed.

Actual work | Unperformed results remain blank

Actual attempt, confidence, observed error, source correction, reviewer objection and changed-context delayed result only if performed. No real learner or reviewer is supplied.

Actual date / case	Attempt / confidence	Exact source / input	Error / original claim	Correction / decision change	Actual objection / response	New delayed context / result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline curves / welfare / arcs / margin	PASS: all baseline formula values independently recalculated, curve points included.
Shock10 with ceiling45 / restore	PASS: shock10/ceiling45 is a coherent changed binding-ceiling case; restored.
Recovery cost0 / restore	PASS: recovery0 restores old-cost equilibrium; worse-than-shock decomposition blocked.
New B700 / restore	PASS: B700 changes weighted arcs and contribution; restored.
New cost170 / restore	PASS: downside new cost170 changes only downside bridge; restored.
Unchanged fare / undefined arcs	PASS: unchanged fare leaves arcs undefined while contribution still computes.
Zero new margin / undefined threshold	PASS: nonpositive new margin leaves positive-margin threshold undefined.
Invalid inputs / independent cases	PASS: twelve invalid input cases preserve independent market/ride model.
Broken welfare reference / restore	PASS: broken welfare total blocks audit/ceiling loss; ride remains independent.
Actual blanks / restored baseline	PASS: all baseline restored, actual attempts/reviewer/delayed evidence blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.